Trauma Emergencies

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Scenario 1: link

38-year-old male | Multisystem Trauma

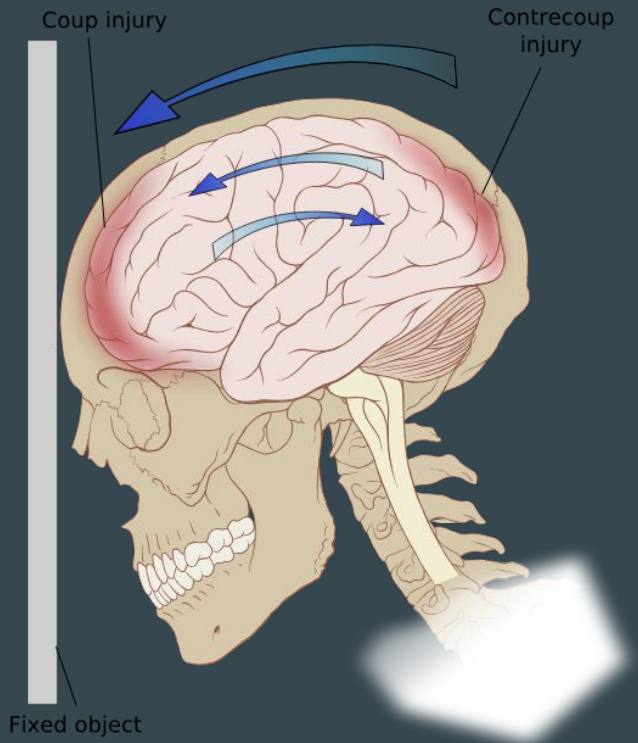
- Altered mental status
- Head & chest trauma
- Bystander called 911

What is the field impression?

Multiple Systems...

1) Traumatic Brain Injury (coup-contrecoup brain injury)
    a) PT signs: Altered Mental Status, Facial Trauma, basal skull fracture
2) Basal Skull Fracture
    a) PT signs: Battle's Signs, Raccoon Eyes, CSF and bleeding in ears (halo test)
3) Spinal Cord Injury / Neurogenic Shock
    a) PT Signs: Very Low BP (<90 sys), slow heart rate, Spinal deformity
4) Flail Chest
    a) PT Signs: Paradoxical Breathing, Bruising on Chest

What is a TBI (specifically, coup-contrecoup)

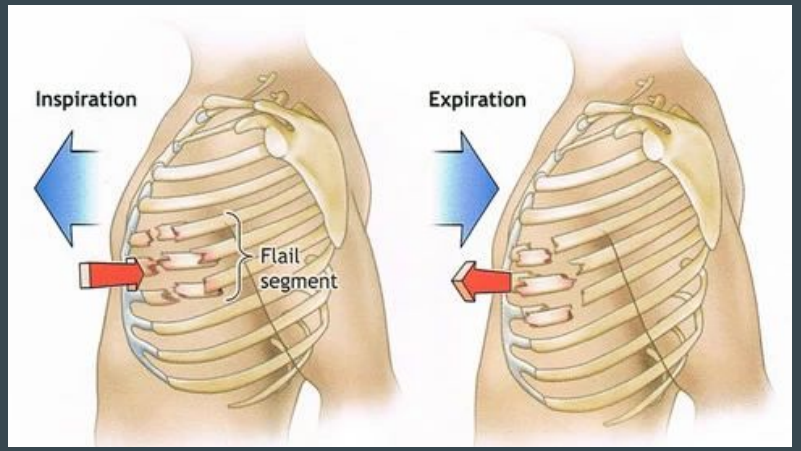


- TBI: Brain injury caused by external force, such as blunt trauma to the head.
- Coup: Injury occurs at the site of impact.
- Contrecoup: Brain movement causes injury on the opposite side of the skull.
- Possible Signs/Symptoms:
  - Head Pain
  - Nausea / Vomiting
  - Balance Issues
  - Noise / Light Sensitivity
  - Altered Mental Status
  - Intracranial Pressure
  - Subdural / Epidural Hematoma
  - Basal Skull Fracture
  - Cervical Spine Fracture
  - Neurogenic Shock
  - Brain Bruising

Some Considerations for Head Trauma

1) Neurological Status
    - Level of consciousness: AVPU, orientation (A&O)
    - Amnesia: Retrograde vs. anterograde
    - Pupillary assessment: PERRL (size, equality, reactivity)
2) Signs of Increased Intracranial Pressure (ICP)
    - Cushing's Triad: Hypertension with widened pulse pressure, bradycardia, and irregular respirations
3) Signs of Skull Fracture
    - Basilar skull fracture: Battle's sign, raccoon eyes, blood or CSF drainage from the nose or ears
    - Suspected CSF leak: Could use halo test
4) External Injuries & Associated Trauma
    - Facial and head assessment: DCAP-BTLS
    - Assess for associated cervical spine injury; maintain spinal precautions when indicated
5) Neurogenic shock
    - Hypotension with relative bradycardia, especially with associated spinal cord injury

What is Flail Chest?



- Caused by Blunt Trauma (Falls, Motor Vehicle Accidents, Assaults)
- Two or more contiguous rib fractures with two or more breakpoints
- Signs/Symptoms:
  - **Paradoxical Breathing**
  - Severe chest pain → shallow breathing
  - Tachycardia
  - Increased RR

Considerations for This Patient

1) Spinal Precautions (C-Spine + Cervical Collar)
2) Extrication: KED board + Backboard
3) Altered Mental Status (A&O x2)
    a) Go through AEIOUTIPS, Head Trauma considerations
4) Flail Chest & Paradoxical Breathing
    a) Administer oxygen via BVM at 15 Lpm
5) Signs of Shock: Skin is pale and diaphoretic, BP: 90/45, low HR
    a) Treat for neurogenic shock (supine position, O_2, blanket)
6) Full-body Assessment reveals spinal deformity, facial trauma, bruising, and minor bleeding at the face.
    a) PT should already be backboarded w/ a cervical collar

ALCO Protocols

TRAUMA PATIENT CARE

- **Routine Medical Care**
- **Critical Interventions** - See below
- **Transport Decision** - Determine need for rapid intervention/transport
- **Transport**
- **If traumatic arrest is suspected do not use ACLS medications**

CRITICAL/TIME SENSITIVE INTERVENTIONS:

- Control major external hemorrhage (see **page 127**)
- Control the Airway - Consider **endotracheal intubation or supraglottic airway device if indicated** (See below for patients with closed head trauma)
- Keep patient warm
- Determine patient severity (see "Trauma Patient Criteria" - see **page 29**):

Meets Physiologic and/or Anatomic Factors	Meets Mechanism of Injury Criteria
→ Transport to the Trauma Center In accordance with Transport Guidelines (page 22). → ADULT - Establish one (1) large bore IV/IO with Normal Saline (NS) or Saline Lock (SL). Establish 2nd IV if appropriate. → PEDIATRIC- Establish one (1) appropriate large bore IV/IO with Normal Saline (NS) or Saline Lock (SL).	→ Transport to the Trauma Center code 2. → ADULT/PEDIATRIC - Establish one (1) large bore IV/IO with Normal Saline (NS) or Saline Lock (SL).
Do NOT delay transport to establish IV/IO access See "Trauma Patient Criteria" (**page 29**) for additional judgment decisions on code 2 transports	

- Consider spinal motion restriction (SMR) for blunt trauma (see **page 139**)
- Administer **Oxygen** - Titrate SpO_2 to 94-99%
- **IV fluid resuscitation:**
  - → SBP < 90 mmHg, NS IV/IO 250 – 500ml bolus
  - → > 90 mmHg, IV/IO TKO or Saline Lock
  - → Reassess BP q 5 minutes
- Consider **TXA** for patients with signs of shock or uncontrolled bleeding (see **page 145**)
- **Care of the patient with a closed head injury** (GCS < 8):
  - → **Advanced airway (ETT or SGA)**
  - → End-tidal CO_2 should be between 30-35 mmHg
  - → Track respirations or ventilate to a rate of approx 12 times/minute with 100% O_2 ***(AVOID HYPERVENTILATION)***
  - → **IV/IO** NS in 500 mL increments to ***maintain mean arterial pressure (MAP) of at least 80 mmHg***. Reassess BP q 5 minutes

IMPORTANT CONSIDERATIONS

- Contact the Base Hospital, if appropriate
- Contact the Trauma Center, as soon as possible
- Consider pain management when appropriate
- Splint fractures and dress wounds ONLY if time permits

FORMULA FOR ESTIMATING MAP

$$\text{MAP} = \text{diastolic} + \frac{(\text{systolic} - \text{diastolic})}{3}$$

BMRC First Aid Protocol

- Unable to obtain OPQRST / SAMPLE (PT is altered)
- Spinal Precautions (hold C-spine)
- Care for shock (provide warmth, place in supine position)
- Closely monitor vitals every 5 minutes
- Call UCPD or 911
- Be ready to give a report to a higher-level-of-care provider.

Senario 2

18yr, female, hurt ankle

What is the field impression?



Sprained ankle

Treatment: splint and transport





Splinting

When to splint?

- Extremity injuries

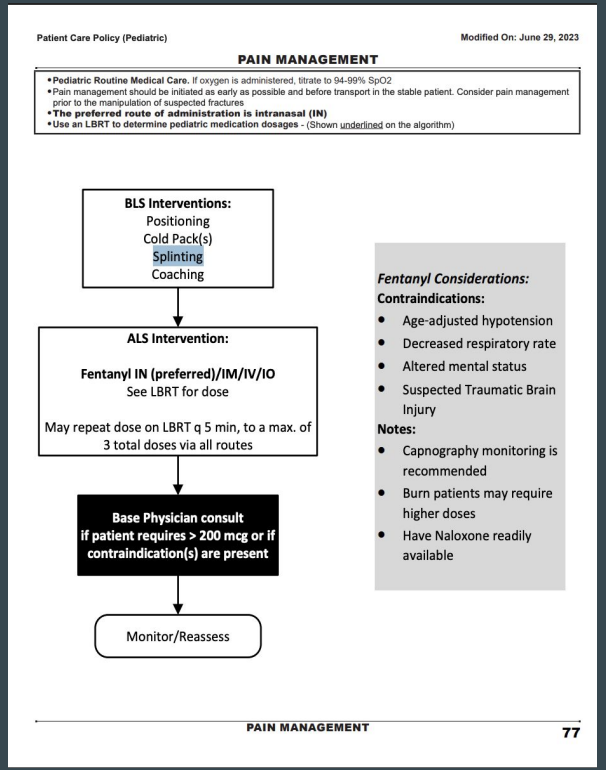
Stop bleeding/cover open wounds with sterile dressing before splinting

- Check CMS before and after
- Stabilize with cardboard, vacuum splint, or SAM splint
- Use swaths to tie to body and prevent movement





ALCO Protocols



Patient Care Policy (Pediatric) Modified On: June 29, 2023

PAIN MANAGEMENT

- **Pediatric Routine Medical Care.** If oxygen is administered, titrate to 94-99% SpO2
- Pain management should be initiated as early as possible and before transport in the stable patient. Consider pain management prior to the manipulation of suspected fractures
- **The preferred route of administration is intranasal (IN)**
- **Use an LBRT to determine pediatric medication dosages** - (Shown underlined on the algorithm)

BLS Interventions:
Positioning
Cold Pack(s)
Splinting
Coaching

ALS Intervention:

Fentanyl IN (preferred)/IM/IV/IO
See LBRT for dose

May repeat dose on LBRT q 5 min, to a max. of 3 total doses via all routes

Base Physician consult if patient requires > 200 mcg or if contraindication(s) are present

Monitor/Reassess

Fentanyl Considerations:

Contraindications:

- Age-adjusted hypotension
- Decreased respiratory rate
- Altered mental status
- Suspected Traumatic Brain Injury

Notes:

- Capnography monitoring is recommended
- Burn patients may require higher doses
- Have Naloxone readily available

PAIN MANAGEMENT 77

▼ **Splints** – cardboard splint with a soft or cushioned surface, flexible, form-fitting splint (e.g., SAM or vacuum splint):			
• Adult Arm	1	1	2
• Adult Leg	1	1	2
• Traction Splint		1	1

Senario 3

24-year-old male | Penetrating Trauma
- Stab wound to right thigh during an altercation
- Significant external bleeding
- Patient complains of severe pain and dizziness
- Bystander called 911

Scene Size-Up & Primary Assessment

Scene Safety

- Scene secured by law enforcement
- PPE and BSI precautions
- One patient; ALS requested

XABCD Assessment

- X: Severe hemorrhage from right thigh
- A: Airway patent
- B: Rapid, shallow breathing (RR 26)
- C: Pale, cool, clammy; rapid, weak pulse

Mild AMS, A&O × 3, GCS 14

Secondary Assessment
Rapid Head-to-Toe
- Head, neck, chest, abdomen, pelvis, back: No obvious injuries
- Right thigh: Deep penetrating wound
- Distal CMS: Assess circulation, motor, sensation

SAMPLE

- S: Severe pain, dizziness, weakness

- A/M/P: NKDA, no medications or significant history

- L: Ate 3 hours ago

- E: Stabbed approximately 10 minutes ago

OPQRST
- Sharp, constant, non-radiating pain
- 9/10 severity
- Worse with movement; slightly relieved by rest

Vital Signs
Blood Pressure 88/54, Heart Rate 132, Respiratory Rate 26, SpO_2 96%, Blood Glucose 105,
GCS 14/15

Clinical Impression: Hemorrhagic Shock
- Hypotension and tachycardia
- Pale, cool, clammy skin
- Mild altered mental status

Trauma Treatment
Immediate Interventions
1. Control bleeding with a tourniquet proximal to the wound.
2. If too proximal for tourniquet placement, use hemostatic wound packing and direct pressure.
3. Reassess bleeding and document tourniquet time.
4. Treat for shock and prevent hypothermia.
5. Monitor airway, breathing, and oxygenation.
6. Request ALS and initiate rapid transport.

BMRC First Aid protocol

Obtain OPQRST and SAMPLE, Head to toe assessment

Care for shock and place patient in a position of comfort

Apply tourniquet or pack wound if needed

Closely monitor vitals every 5 minutes

Call UCPD or 911

Be ready to give a report to a higher-level-of-care provider.

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Thank you + questions

Kahoot!